Introduction to Canonical Correlation Analysis

Canonical Correlation Analysis (CCA) is a multivariate statistical method that seeks to understand the relationships between two sets of variables by finding the linear combinations of variables in each set that are maximally correlated. To build up an understanding of CCA, we begin with fundamental concepts like covariance and correlation and gradually work toward the derivation of CCA.

1 Covariance Between Two Random Variables

Given two random variables X and Y of real numbers, the **covariance** measures how much the two variables vary together. It is defined as:

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

- When $\mathrm{Cov}(X, Y) > 0$, it means that X and Y tend to increase together (positive relationship). - When $\mathrm{Cov}(X, Y) < 0$, it means that X and Y tend to vary in opposite directions (negative relationship).

If we have two vectors of observations $x = [x_1, x_2, \dots, x_n]$ and $y = [y_1, y_2, \dots, y_n]$, we can compute the sample covariance:

$$\mathrm{Cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

where $\bar{x}$ and $\bar{y}$ are the sample means of x and y, respectively.

In summary, dividing by $n - 1$ when calculating sample covariance helps to obtain an unbiased estimator of the true population covariance. This practice is standard in statistics and helps ensure that our estimators reflect the true relationships between the variables in the population.

2 Covariance Matrix

Given a random vector $X \in \mathbb{R}^p$, the **covariance matrix** $\Sigma \in \mathbb{R}^{p \times p}$ describes the covariances between each pair of components of X. If $X = [X_1, X_2, \dots, X_p]^T$, then the covariance matrix Σ is given by:

$$\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$$

In other words, Σ is symmetric, and its diagonal entries represent the variances of the individual variables, while the off-diagonal entries represent the covariances between different variables.

A data matrix $X \in \mathbb{R}^{n \times p}$ represents a collection of observations where each row corresponds to an observation of a random vector x and each column corresponds to a specific feature across all observations. **We often assume that these observations are realizations from a joint distribution**, with each feature modeled as a random variable, allowing us to analyze statistical properties like means, variances, and covariances. This framework is fundamental in statistics and machine learning, as it facilitates understanding relationships among variables and guides data analysis and modeling methods.

3 Correlation Between Two Random Variables

The **correlation** between two random variables X and Y normalizes the covariance to obtain a measure that ranges between -1 and 1. The correlation is defined as:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

Where $\text{Var}(X)$ and $\text{Var}(Y)$ are the variances of X and Y.

If X and Y are vectors, the correlation between X and Y is calculated in the same manner, with the covariance and variance taken over the elements of the vectors.

4 Generalization to Canonical Correlation Analysis

Let $X \in \mathbb{R}^{n \times p}$ and $Y \in \mathbb{R}^{n \times q}$ be two datasets, where each row of X and Y corresponds to an observation, and each column represents a variable.

The goal of **Canonical Correlation Analysis (CCA)** is to find linear combinations of the columns of X and Y such that the correlation between these linear combinations is maximized.

Let $U = Xa$ and $V = Yb$, where $a \in \mathbb{R}^p$ and $b \in \mathbb{R}^q$ are the vectors of canonical coefficients for X and Y, respectively. The objective is to maximize the correlation between U and V:

$$\rho(a, b) = \frac{\text{Cov}(U, V)}{\sqrt{\text{Var}(U)}\sqrt{\text{Var}(V)}}$$

5 Covariance and Correlation for Canonical Variables

Given $U = Xa$ and $V = Yb$, we have:

$$\begin{aligned}\mathrm{Cov}(U, V) &= a^T S_{XY} b \\ \mathrm{Var}(U) &= a^T S_{XX} a \\ \mathrm{Var}(V) &= b^T S_{YY} b\end{aligned}$$

Where S_{XY} is the cross-covariance matrix between X and Y:

$$S_{XY} = \frac{1}{n-1} X^T Y$$

And S_{XX} and S_{YY} are the covariance matrices of X and Y, respectively:

$$S_{XX} = \frac{1}{n-1} X^T X, \quad S_{YY} = \frac{1}{n-1} Y^T Y$$

6 Canonical Correlation Optimization Problem

The objective of CCA is to find the vectors $a \in \mathbb{R}^p$ and $b \in \mathbb{R}^q$ that maximize the correlation $\rho(a, b)$:

$$\rho(a, b) = \frac{a^T S_{XY} b}{\sqrt{a^T S_{XX} a}\sqrt{b^T S_{YY} b}}$$

Reformulating as a Constrained Optimization Problem Instead of working with the ratio directly, we maximize the numerator $a^T S_{XY} b$ subject to the normalization constraints that ensure the variances of U and V are 1:

$$\begin{aligned}&\max_{a,b} \quad a^T S_{XY} b \\ &\text{subject to} \quad a^T S_{XX} a = 1 \quad \text{and} \quad b^T S_{YY} b = 1\end{aligned}$$

Given the correlation between two linear combinations $U = Xa$ and $V = Yb$, we want to maximize the following ratio:

$$\rho = \frac{a^T S_{XY} b}{\sqrt{a^T S_{XX} a}\sqrt{b^T S_{YY} b}}.$$

7 Using Homogeneity

Notice that if we scale a and b by a common factor λ, the correlation remains unchanged:

$$\rho(\lambda a, \lambda b) = \frac{(\lambda a)^T S_{XY} (\lambda b)}{\sqrt{(\lambda a)^T S_{XX} (\lambda a)}\sqrt{(\lambda b)^T S_{YY} (\lambda b)}}.$$

This can be simplified as follows:

$$\rho(\lambda a, \lambda b) = \frac{\lambda^2 a^T S_{XY} b}{\lambda^2 \sqrt{a^T S_{XX} a}\sqrt{b^T S_{YY} b}} = \rho(a, b).$$

This property indicates that the problem is homogeneous. Hence, we can impose a constraint on a and b without loss of generality, which simplifies our maximization problem.

8 Why are These Problems Equivalent?

The first problem involves maximizing a ratio, which is often more complex due to the dependencies between the numerator and the denominator. By introducing constraints that normalize the variances of the canonical variables, we effectively fix the denominator, allowing us to focus solely on maximizing the numerator. This reformulation simplifies the optimization problem and ensures that we find the same optimal solution as we would by maximizing the original ratio.

9 Solving the Optimization Problem: Lagrangian and Generalized Eigenvalue Problem

To solve this constrained optimization problem, we use Lagrange multipliers to incorporate the constraints into the objective function.

Define the Lagrangian:

$$\mathcal{L}(a, b, \lambda_a, \lambda_b) = a^T S_{XY} b - \frac{\lambda_a}{2}(a^T S_{XX} a - 1) - \frac{\lambda_b}{2}(b^T S_{YY} b - 1)$$

Take the derivative of $\mathcal{L}$ with respect to a and set it equal to zero:

$$\frac{\partial \mathcal{L}}{\partial a} = S_{XY} b - \lambda_a S_{XX} a = 0$$

Rearranging gives:

$$S_{XY} b = \lambda_a S_{XX} a$$

Similarly, take the derivative with respect to b and set it equal to zero:

$$\frac{\partial \mathcal{L}}{\partial b} = S_{YX} a - \lambda_b S_{YY} b = 0$$

Rearranging gives:

$$S_{YX} a = \lambda_b S_{YY} b$$

10 Generalized Eigenvalue Problem

We now derive the generalized eigenvalue problem. From $S_{XY}b = \lambda_a S_{XX}a$, substitute b from $S_{YX}a = \lambda_b S_{YY}b$:

$$S_{XY}S_{YY}^{-1}S_{YX}a = \lambda_a^2 S_{XX}a$$

This is the generalized eigenvalue problem:

$$(S_{XX}^{-1}S_{XY}S_{YY}^{-1}S_{YX})a = \lambda^2 a$$

where λ^2 are the squared canonical correlations.

Similarly, the generalized eigenvalue problem for b is:

$$(S_{YY}^{-1}S_{YX}S_{XX}^{-1}S_{XY})b = \lambda^2 b$$

11 Conclusion

Canonical Correlation Analysis (CCA) finds linear combinations of the variables in two datasets X and Y such that the correlation between these linear combinations is maximized. Through the optimization of the correlation function, we arrive at a generalized eigenvalue problem that can be solved to obtain the canonical correlations and their corresponding canonical coefficients.

This approach highlights the interconnectedness between covariance, correlation, and canonical correlations, providing a robust statistical method for analyzing the relationships between multivariate datasets.

12 Canonical Variables and Weights in CCA

In Canonical Correlation Analysis (CCA), we derive new variables known as canonical variables from two datasets. The relationships and terminologies are as follows:

1. **Canonical Variables**: - The linear combinations formed from the original variables in each dataset are referred to as canonical variables. - For Dataset X, the canonical variable is denoted as U. - For Dataset Y, the canonical variable is denoted as V.

2. **Weights or Coefficients**: - These coefficients are used to create the canonical variables from the original variables. - The weights a are utilized to combine the original variables from Dataset X to create the canonical variable U:

$$U = Xa.$$

- The weights b are used to combine the original variables from Dataset Y to create the canonical variable V:

$$V = Yb.$$

12.1 Example

To illustrate this concept, consider the following example: - Let Dataset X consist of features such as Math, English, and Science scores. - Let Dataset Y consist of features like overall performance ratings and participation scores.

After performing CCA, we might derive: - The canonical variable U as:

$$U = 0.5 \times \text{Math} + 0.3 \times \text{English} + 0.2 \times \text{Science},$$

where the weights a represent the contribution of each feature in Dataset X.

- The canonical variable V can be expressed as:

$$V = 0.6 \times \text{Performance} + 0.4 \times \text{Participation},$$

where the weights b indicate the contribution of each feature in Dataset Y.

In this example: - U and V are the canonical variables that summarize the information from the original datasets. - The coefficients a and b reveal how much influence each original variable has in forming these canonical variables.

12.2 Analysis of Math Scores and Performance Ratings

In the context of the provided example, the relationship between Math scores (from Dataset X) and performance ratings (from Dataset Y) can be analyzed as follows:

- The coefficients derived from the canonical variate for Math indicate that Math scores significantly contribute to the formation of the canonical variate U. In our example, the coefficient of 0.5 for Math suggests that it plays an important role in the overall relationship with performance.

- The canonical correlation between the first canonical variate U and the first canonical variate V quantifies the relationship between these linear combinations. A high canonical correlation indicates that as Math scores increase, performance ratings also tend to increase.

- This analysis suggests that improving Math scores could positively affect overall performance ratings. If CCA reveals that Math scores are influential in determining performance, targeted interventions in math education could yield significant improvements.

In summary, the analysis shows that while U and V are derived from their respective datasets, the canonical correlation and the weights highlight the interdependence of Math scores and performance ratings, providing insights for educational focus and improvement.